

This is the beginning prompt:

i will provide you with the homework question and the output of the cell, help me to answer the questions

I then provide the cell's output and problem set question:

Here is the generated response:

Overall, ChatGPT performed strongly in completing the Visualizing BERT and GPT task, accurately describing the attention patterns in both models and consistently aligning with the staff solutions. Its responses correctly identified key concepts such as GPT's sequential autoregressive attention, BERT's bidirectionality, and contextual disambiguation of polysemous words like "play." INo hallucinations were observed, and although the task was not heavily coding-oriented, ChatGPT still demonstrated an understanding of the code's purpose and provided valid interpretations of the outputs.

The main areas for improvement involved adding specific details emphasized in the staff's answer. These omissions were minor in nature and related more to phrasing than conceptual understanding, showing that ChatGPT's reasoning was correct overall and only required slight alignment with expected instructional emphasis.

Homework Write-Up: Visualizing Attention in GPT-2 and BERT

(a) Attention in GPT-2

i. Similarities & Differences Across Examples

Across the three examples, GPT consistently attends most strongly to **nearby/contextually related tokens**, but the complexity of patterns increases with sentence length.

- In “**The dog ran**”, attention is symmetric and simple; each word mainly attends to its neighbors (see page 3).
- In “**The dog sitting in the car**”, attention becomes denser as modifiers (“sitting”, “in”, “car”) depend on the noun phrase (see page 3). These modifier words receive stronger attention.

Successfully point out the some local attention in “The dog ran” and “The dog sitting in the car”. But missing to point out an important concept that GPT pays more attention to dog rather than car.

- In the long sentence about **Jaime Salazar**, attention spreads further across the sentence, forming many long-range lines linking the name to pronouns and later mentions (“his”, “Mr.”) (page 3).
→ This shows GPT-2 attempting to maintain global context (entity linking) across multiple clauses.

👉 **Observation:** Queries for nouns (like *Salazar*) link to later pronouns, while verbs attend to nearby nouns for tense consistency.

ii. Attention Across GPT Layers

Changing layers produces different patterns:

- **Lower layers** (Layer 0 shown) mostly encode local structure by attending to adjacent tokens (simple zig-zag).
- **Higher layers** (when toggled) tend to show more connections to semantically related tokens and long-distance dependencies (e.g., name to pronoun, repeated entity).

👉 **Conclusion:** GPT shifts from **local syntax** to **global semantics** as depth increases.

For me, I would also say GPT pays many attention to the first token. GPT5.1 also makes a point in it's response.

(b) BERT Pays Attention

iii. BERT Layer Patterns

In the example (“I wanted to have a party” + “I bought a cake for it”), attention highlights:

- **Pronouns strongly attend to antecedents** (“it” → “party”) (page 4).
- **[CLS]** attends broadly across the sentence, indicating sentence-level summarization.
- **Function words (to, a, for)** produce weaker or scattered patterns, verifying that BERT learns to ignore low-information tokens.

👉 BERT allocates attention based on **semantic relevance**, not just proximity.

Some valid point was made about the CLS token and the pronouns, aliens with the provided graph

iv. GPT vs. BERT Attention

- **GPT is autoregressive**; attention primarily flows **backward** to previous words (cannot see the future).
- **BERT is bidirectional**, attending to both earlier and later tokens simultaneously. In the example, BERT lets “it” attend forward and backward to clues in both sentences (page 4).

👉 **Result:** BERT better resolves pronouns and sentence relations, while GPT excels at sequential language modeling.

Aliens fully with the staff’s response most about the GPT’s autoregressive and BERT’s bidirectional

v. Same Structure, Different Meaning (“play” example)

The word “**play**” exhibits distinct attention patterns in:

- “*to play at the park*” (verb meaning activity)
- “*a play in the park*” (noun meaning performance)

On page 5, attention differs:

- In Sentence A, “**play**” **attends to “to”** (verb marker).
- In Sentence B, “**play**” **attends to “a”**, indicating a noun.

👉 This shows **BERT learns distinct embeddings for polysemous words**, purely from contextual cues.

The explanation correctly identified contextual differences in the word “play,” but would match the staff framing more closely by specifying that these differences appear in the keys rather than in the overall pattern.

vi. Bidirectionality in Reverse Input

When sentence order is reversed (page 6), BERT still:

- Links pronouns correctly
- Tracks dependencies in both directions
- Preserves entity referencing

👉 Even with reversed input, BERT maintains contextual comprehension → clear evidence of **bidirectional encoding**.

The discussion of bidirectionality was valid, but should explicitly reference that this ability comes from large-scale self-supervised pretraining, as emphasized by the staff response.

vii. Effect of Pre-Training

Pre-training enables:

- Robust entity linking (**it** ↔ **party**)
- Disambiguation of polysemous words (**play**)
- Attention suppression on non-informative tokens

Without massive pre-training, these semantic distinctions would not emerge.

(c) Multiple Heads

viii. Features Across Heads

From page 8 grid:

- Some heads strongly highlight **local syntax** (next-token relationships).

- Others focus on **long-range semantic links**.
- Others emphasize **punctuation**, **[CLS]**, **[SEP]**, or sentence boundaries.

👉 **Different heads specialize**, allowing the model to learn many linguistic patterns concurrently.

The description of head specialization was accurate,

(d) Random vs Trained Attention

x. Differences After Random Initialization

Randomized BERT shows:

- chaotic attention with **no consistent links**
- loss of entity tracking
- weak and uniform line patterns

Trained BERT shows **structured, meaningful clustering**, especially around pronouns, entities, and sentence markers.

👉 **Training transforms attention from noise into linguistic structure.**

The identified features were correct as well, aligns with the provided graph

xi. Expected Strong Gradients

Even without training, high-gradient relationships would likely include:

- **Pronouns ↔ nouns**
- **Verbs ↔ subjects**
- **[CLS] ↔ all tokens** (global pooling)
- **Polysemous words ↔ disambiguating neighbors**

Because these tokens **maximize loss reduction**, the model should learn them first when training begins.

The attention gradients were characterized appropriately, but should note that the strongest expected connections include neighbors and separator tokens, to consistent with the staff explanation.

If you'd like, I can also:

- ◆ Export these answers into a PDF
- ◆ Shorten to bullet-point form for submission
- ◆ Create visual annotations on screenshots

Just let me know what you prefer!